

Fractional Factorial Design 2^{4-1}

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Introduction

We are going to project a 4 factor factorial design (A,B,C,D) to 3 factor fractional design of scale 4

The Data

First thing's first, we insert the data. All of them.

```
Y<- c(45,71,48,65,68,60,80,65,43,100,45,104,75,86,70,96)
#(1),a,b,ab,c,ac,bc,abc,d,ad,bd,abd,cd,acd,bcd,abcd
```

We are going to create the design matrix as usual:

```
D <- rep(c(-1,1), each = 8)
C <- rep(rep(c(-1,1), each = 4), 2)
B <- rep(rep(c(-1,1), each = 2), 4)
A <- rep(rep(c(-1,1), each = 1), 8)
AB <- A*B
AC <- A*C
AD <- A*D
BC <- C*B
BD <- B*D
CD <- C*D
ABC <- A*C*B
ABD <- A*B*D
ACD <- A*C*D
BCD <- B*C*D
ABCD<- A*B*C*D
#We create the design matrix
Design <- data.frame(
```

```

Y    = Y,
Intercept = 1,
A    = A,
B    = B,
C    = C,
D    = D,
AB   = AB,
AC   = AC,
AD   = AD,
BC   = BC,
BD   = BD,
CD   = CD,
ABC  = ABC,
ABD  = ABD,
ACD  = ACD,
BCD  = BCD,
ABCD= ABCD
)
Design

```

	Y	Intercept	A	B	C	D	AB	AC	AD	BC	BD	CD	ABC	ABD	ACD	BCD	ABCD
1	45		1	-1	-1	-1	-1	1	1	1	1	1	-1	-1	-1	-1	1
2	71		1	1	-1	-1	-1	-1	-1	1	1	1	1	1	1	-1	-1
3	48		1	-1	1	-1	-1	1	1	-1	-1	1	1	1	-1	1	-1
4	65		1	1	1	-1	-1	1	-1	-1	-1	1	-1	-1	1	1	1
5	68		1	-1	-1	1	-1	1	-1	1	-1	1	1	-1	1	1	-1
6	60		1	1	-1	1	-1	-1	1	-1	1	-1	-1	1	-1	1	1
7	80		1	-1	1	1	-1	-1	1	1	-1	-1	-1	1	1	-1	1
8	65		1	1	1	1	-1	1	1	-1	1	-1	-1	1	-1	-1	-1
9	43		1	-1	-1	-1	1	1	1	-1	1	-1	-1	1	1	1	-1
10	100		1	1	-1	-1	1	-1	-1	1	1	-1	-1	1	-1	1	1
11	45		1	-1	1	-1	1	-1	1	-1	1	-1	1	-1	1	-1	1
12	104		1	1	1	-1	1	1	-1	1	-1	1	-1	1	-1	-1	-1
13	75		1	-1	-1	1	1	1	-1	-1	-1	1	1	1	-1	-1	1
14	86		1	1	-1	1	1	-1	1	1	-1	-1	1	-1	1	-1	-1
15	70		1	-1	1	1	1	-1	-1	1	1	1	-1	-1	-1	1	-1
16	96		1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

We are going to choose the lines of this design matrix according to the generator I=ABCD, or D=ABC:

```
Design_FFD2 <- subset(Design, D == ABC)
```

```
Design_FFD2
```

	Y	Intercept	A	B	C	D	AB	AC	AD	BC	BD	CD	ABC	ABD	ACD	BCD	ABCD
1	45		1	-1	-1	-1	-1	1	1	1	1	1	-1	-1	-1	-1	1
4	65		1	1	1	-1	-1	1	-1	-1	-1	-1	1	-1	1	1	1
6	60		1	1	-1	1	-1	-1	1	-1	-1	1	-1	1	-1	1	1
7	80		1	-1	1	1	-1	-1	1	1	-1	-1	-1	1	1	-1	1
10	100		1	1	-1	-1	1	-1	-1	1	1	-1	-1	1	-1	1	1
11	45		1	-1	1	-1	1	-1	1	-1	-1	1	-1	1	1	-1	1
13	75		1	-1	-1	1	1	1	-1	-1	-1	-1	1	1	-1	-1	1
16	96		1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Now I run the ANOVA:

```
# We turn our data into a data frame
df2 <- data.frame(
  Y = Design_FFD2$Y,
  A = Design_FFD2$A,
  B = Design_FFD2$B,
  C = Design_FFD2$C,
  D = Design_FFD2$D
)

model2 <- lm(Y ~ A*B*C*D, data = df2)
summary(model2)
```

Call:

```
lm(formula = Y ~ A * B * C * D, data = df2)
```

Residuals:

ALL 8 residuals are 0: no residual degrees of freedom!

Coefficients: (8 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	70.75	NaN	NaN	NaN
A	9.50	NaN	NaN	NaN
B	0.75	NaN	NaN	NaN
C	7.00	NaN	NaN	NaN

D	8.25	NaN	NaN	NaN
A:B	-0.50	NaN	NaN	NaN
A:C	-9.25	NaN	NaN	NaN
B:C	9.50	NaN	NaN	NaN
A:D	NA	NA	NA	NA
B:D	NA	NA	NA	NA
C:D	NA	NA	NA	NA
A:B:C	NA	NA	NA	NA
A:B:D	NA	NA	NA	NA
A:C:D	NA	NA	NA	NA
B:C:D	NA	NA	NA	NA
A:B:C:D	NA	NA	NA	NA

Residual standard error: NaN on 0 degrees of freedom
Multiple R-squared: 1, Adjusted R-squared: NaN
F-statistic: NaN on 7 and 0 DF, p-value: NA

It is clear that we can only calculate all the main effects and 3 of the 6 2-factor interactions.

Now we want to display the estimates again but in a column from the biggest absolute value to the lowest:

```
# Get the estimated coefficients (exclude the intercept)
effects <- coef(model2)[-1]

# Sort by absolute value, descending
effects_sorted <- effects[order(-abs(effects))]

# Display
effects_sorted
```

A	B:C	A:C	D	C	B	A:B	A:D	B:D	C:D
9.50	9.50	-9.25	8.25	7.00	0.75	-0.50	NA	NA	NA
A:B:C	A:B:D	A:C:D	B:C:D	A:B:C:D					
NA	NA	NA	NA	NA					

We can clearly see that A is important, as well as C and D. We can also see that BC is important but BC is the same with AD which is the dominant one as both A and D are more important than B and C. AC is also important.

We can now do the same thing but with the opposite generator $I = -ABCD$, or $D = -ABC$.

```
Design_FFD <- subset(Design, D == -ABC)
```

```
Design_FFD
```

	Y	Intercept	A	B	C	D	AB	AC	AD	BC	BD	CD	ABC	ABD	ACD	BCD	ABCD
2	71	1	1	-1	-1	-1	-1	-1	-1	1	1	1	1	1	1	-1	-1
3	48	1	-1	1	-1	-1	-1	1	1	-1	-1	1	1	1	-1	1	-1
5	68	1	-1	-1	1	-1	1	-1	1	-1	1	-1	1	-1	1	1	-1
8	65	1	1	1	1	-1	1	1	-1	1	-1	-1	1	-1	-1	-1	-1
9	43	1	-1	-1	-1	1	1	1	-1	1	-1	-1	-1	1	1	1	-1
12	104	1	1	1	-1	1	1	-1	1	-1	1	-1	-1	1	-1	-1	-1
14	86	1	1	-1	1	1	-1	1	1	-1	-1	1	-1	-1	1	-1	-1
15	70	1	-1	1	1	1	-1	-1	-1	1	1	1	-1	-1	-1	1	-1

Now I run the ANOVA again but for these other 8 responses:

```
# We turn our data into a data frame
df <- data.frame(
  Y = Design_FFD$Y,
  A = Design_FFD$A,
  B = Design_FFD$B,
  C = Design_FFD$C,
  D = Design_FFD$D
)

model <- lm(Y ~ A*B*C*D, data = df)
summary(model)
```

Call:

```
lm(formula = Y ~ A * B * C * D, data = df)
```

Residuals:

ALL 8 residuals are 0: no residual degrees of freedom!

Coefficients: (8 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	69.375	NaN	NaN	NaN
A	12.125	NaN	NaN	NaN
B	2.375	NaN	NaN	NaN
C	2.875	NaN	NaN	NaN

D	6.375	NaN	NaN	NaN
A:B	0.625	NaN	NaN	NaN
A:C	-8.875	NaN	NaN	NaN
B:C	-7.125	NaN	NaN	NaN
A:D	NA	NA	NA	NA
B:D	NA	NA	NA	NA
C:D	NA	NA	NA	NA
A:B:C	NA	NA	NA	NA
A:B:D	NA	NA	NA	NA
A:C:D	NA	NA	NA	NA
B:C:D	NA	NA	NA	NA
A:B:C:D	NA	NA	NA	NA

Residual standard error: NaN on 0 degrees of freedom
Multiple R-squared: 1, Adjusted R-squared: NaN
F-statistic: NaN on 7 and 0 DF, p-value: NA

It is clear once again that we can only calculate all the main effects and 3 of the 6 2-factor interactions.

Now we want to display the estimates again from the biggest absolute value to the lowest:

```
# Get the estimated coefficients (exclude the intercept)
effects <- coef(model)[-1]

# Sort by absolute value, descending
effects_sorted <- effects[order(-abs(effects))]

# Display
effects_sorted
```

A	A:C	B:C	D	C	B	A:B	A:D	B:D	C:D
12.125	-8.875	-7.125	6.375	2.875	2.375	0.625	NA	NA	NA
A:B:C	A:B:D	A:C:D	B:C:D	A:B:C:D					
NA	NA	NA	NA	NA					

Once again it is visible that the main effects A,C,D as well as the interactions AC, AD are the important ones.

To finish we are going to calculate a Pseudo-Standard-Error(PSE) to calculate estimated t-values to be certain about what is important and what is not:

	Factor	Estimate	PSE	t_value
A	A	9.50	12.375	0.76767677
B:C	B:C	9.50	12.375	0.76767677
A:C	A:C	-9.25	12.375	-0.74747475
A:B:C	A:B:C	8.25	12.375	0.66666667
C	C	7.00	12.375	0.56565657
B	B	0.75	12.375	0.06060606
A:B	A:B	-0.50	12.375	-0.04040404

Normal Probability Plot of Effects (Lenth)

